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Module 7: Assignment - Security-usability design choices

1. Robust security and convenience of use must be balanced in the MCY Fast Food online ordering system. When it comes to usability, customers anticipate a seamless experience when they browse menus, add or remove things from their basket, enter personal information, pay, and receive confirmations. In addition to being mobile-friendly and straightforward, the interface should steer clear of pointless procedures and needless data entering. While delays, unclear navigation, or repeated authentication requests can lead to dissatisfaction and order abandonment, features like responsive buttons, rapid confirmation emails, and explicit pricing help create a positive user experience.
Strong countermeasures that are invisible to the user must be incorporated into the standard workflow from a security-usability standpoint. Data interception, session hijacking, cart manipulation, and insider exploitation of consumer information are among the main threats. Without making things more complicated for patrons, mitigations including PCI DSS compliant payment gateways, secure session handling, HTTPS encryption, and access controls for restaurant employees should be put into place. The system should apply a practical security architecture that safeguards sensitive data while preserving an effective, user-friendly ordering procedure, as opposed to striving for "perfect" security that can impair usability.

2. A number of probable demotivators could cause consumers to inadvertently find a way around or compromise security measures on the MCY Fast Food online ordering system. Customers may stop using secure payment channels and switch to less secure ones (such as calling the store and orally providing card details) if the payment procedure is lengthy, necessitates several re-authentications, or rejects legitimate cards without providing clear justifications. Similar to this, customers may become frustrated by excessively stringent password or account creation restrictions during checkout, which may lead them to reuse weak or popular passwords and raise the possibility of credential breach.
Unnecessary or excessively frequent security reminders are another typical demotivator. For instance, requiring clients to frequently complete captchas or reenter payment information may cause them to disable browser security settings or save private data insecurely for convenience. Lack of clear feedback or error signals may also encourage users to refresh pages or click buttons multiple times, potentially duplicating orders or exposing payment sessions to tampering. These situations demonstrate how weakly balanced security controls may inadvertently encourage users to engage in risky activities that compromise the system's overall security.

3. The fact that secure procedures are notably more complicated than standard, insecure alternatives is a major motivation for risky behavior in the MCY Fast Food ordering system. U sers can prefer quicker but less secure choices, such keeping credit card information in the browser or utilizing an unencrypted payment link, if the secure payment gateway demands several redirections, additional logins, or drawn out authentication stages. Similar to this, if registering an account with a strong password is required before placing an order, some consumers may choose to use weak, simple passwords or share one account with several persons, which could lead to vulnerabilities.
The disruption of the planned workflow caused by security restrictions serves as another motive. Customers may attempt to get around authentication by leaving sessions open on shared or public devices if they are frequently required to reenter data after being suddenly locked out because of session timeouts.

Customers may call the restaurant directly and expose private order and payment information through unprotected means if confirmation emails are not sent promptly or contain insufficient information. In each of these situations, users are influenced to make decisions that compromise overall security because they believe that secure procedures are slower or more complicated than typical usage circumstances.

4. Triggers for the MCY Fast Food ordering platform should automatically safeguard users by identifying unsafe actions, such as repeated unsuccessful payments, extended periods of inactivity during checkout, or odd order patterns, and initiating session timeouts or verification. Clear and actionable warnings that notify users of unsafe connections, missing or suspicious information, or terminating sessions must be given without being overbearing. Lock symbols, "Secure Payment" labels, progress bars, and branded confirmation messages with order data are examples of indicators that should continuously reassure users during the ordering process.